Oleksandr Nekrashevych

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Education

University of Pretoria Expected Graduation: December 2026
BEng Computer Engineering *Pretoria, Gauteng*

- **Relevant Coursework:** Control Systems, Signal Processing, State Estimation (KF/EKF/UKF), Machine Learning, Computer Vision, Robotics, Embedded Systems, Algorithms

Projects

Facial Feature Extraction & Robotic Sketching System | *Python, C, Computer Vision, Embedded Systems*

- Developed an end-to-end computer vision system that extracts facial structure from images and reconstructs simplified sketches via a custom 3-axis robotic drawing platform.
- Implemented a first-principles feature extraction pipeline using Histogram of Oriented Gradients (HOG), edge detection, and spatial localization to isolate key facial regions.
- Built embedded CNC control firmware translating image-derived features into coordinate-based motion commands for automated sketch generation.

Vision-Based Autonomous Chess Agent | *Python, Computer Vision, Automation*

- Built an autonomous chess-playing system that operates directly on a graphical chess interface using real-time computer vision, engine-based decision-making, and system-level input automation.
- Implemented screen-based board detection and piece state extraction using OpenCV to reconstruct game state from raw visual input without APIs or scraping.
- Integrated Stockfish for move generation and developed automated mouse control to execute moves, forming a closed perception–decision–action loop.

Multi-Colour Line-Following Robot (Robot Race Day – Semi-Finalist) | *Control Systems, Signal Processing, Embedded Systems*

- Developed an autonomous embedded robotics system capable of tracking multiple coloured race lines using real-time photodiode sensing and adaptive control logic.
- Implemented low-level assembly firmware on a PIC-based microcontroller for deterministic motor control and real-time sensor processing in dynamic environments.
- Designed a first-principles analog signal processing pipeline using custom photodiode circuitry for robust multi-colour line detection without external libraries.
- Achieved semi-finalist placement in the 2024 TUKS Robot Race Day through reliable autonomous navigation under varying track conditions.

Experience

Cheery Robot Academy | *Programming & Robotics Instructor* April 2024 – Present

- Teach programming and robotics across middle school, high school, and university levels, covering software engineering fundamentals and embedded systems.
- Deliver instruction in Java, Python, and C++, alongside applied development in Roblox Studio and Godot.
- Run hands-on robotics and electronics sessions focused on microcontroller systems, sensors, and embedded systems design.
- Supervise student projects in game development, automation, and interactive systems with real-world engineering focus.

Upwork | *Freelance Software Developer* July 2024 – July 2025

- Designed 80+ SQL exercises for a PostgreSQL Udemy course used by 5,000+ learners, covering joins, indexing, constraints, subqueries, and query optimisation.
- Structured curriculum flow and refined explanations to improve clarity and accessibility for beginner to intermediate learners.
- Collaborated with the course creator to ensure instructional consistency and delivered production-ready content, achieving a 5-star rating.

RunJumpyFly | *Frontend & Embedded Systems Intern* July 2024 – August 2024

- Built a responsive financial consultancy website section from Figma designs using Webflow, HTML, CSS, and JavaScript.
- Developed reusable UI components and interactive elements to improve consistency and cross-device responsiveness.
- Implemented microcontroller-based LED strip control logic enabling programmable lighting sequences for embedded systems experimentation.

KeyStone Electronic Solutions | *Backend Vacation Work Intern* July 2022

- Worked on a legacy Java Spring Boot system, resolving dependency and version conflicts on modern Ubuntu environments.
- Assisted in integrating MySQL for persistent data storage and backend data retrieval.
- Containerised the application using Docker and aligned dependencies for stable and reproducible deployment.

Certifications

Cisco Networking Academy December 2020 – January 2024
Cisco Networking Certifications *Pretoria, Gauteng*

- CCNA: Introduction to Networks
- CCNA: Routing and Switching Essentials
- CCNAv7: Switching, Routing, and Wireless Essentials
- CCNAv7: Enterprise Networking, Security and Automation
- CCNP Enterprise: Core Networking

Technical Skills

Languages: Python, C, C++, Java, JavaScript, Lua, Assembly
Technologies: Node.js, Docker, Git, MySQL, PostgreSQL, MongoDB, OpenCV, MATLAB, LTspice, HTML, CSS, Webflow, Roblox Studio, Godot
Concepts: Software Architecture, Computer Vision, Robotics, Autonomous Systems, Control Systems, State Estimation (KF/EKF/UKF), Real-Time Systems, Signal Processing, Embedded Systems